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Bernstein–Szegő–Lebesgue Sobolev Orthogonal Polynomials on the Unit Circle

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In this paper we consider the following continuous Sobolev inner product on the unit circle

$$\langle f(z), g(z) \rangle_s = \int_0^{2\pi} f(e^{i\theta}) \overline{g(e^{i\theta})} d\mu(\theta) + \frac{1}{\lambda} \int_0^{2\pi} f'(e^{i\theta}) \overline{g'(e^{i\theta})} \frac{d\theta}{2\pi}, \quad z = e^{i\theta},$$

where $\lambda > 0$, $d\mu(\theta)$ is a Bernstein–Szegő measure and $d\theta/2\pi$ is the normalized Lebesgue measure on $[0, 2\pi]$. We study the corresponding sequence of orthogonal polynomials. Algebraic properties, asymptotic behavior and asymptotic distribution of zeros for such polynomials are obtained.

Keywords: Orthogonal polynomials; Difference equations;
Sobolev inner products; Measures on the unit circle

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1 INTRODUCTION

The aim of this paper is to study algebraic and analytic properties of polynomials orthogonal with respect to the Sobolev inner product

$$\langle f(z), g(z) \rangle_s = \int_0^{2\pi} f(e^{i\theta}) \overline{g(e^{i\theta})} d\mu(\theta) + \frac{1}{\lambda} \int_0^{2\pi} f'(e^{i\theta}) \overline{g'(e^{i\theta})} \frac{d\theta}{2\pi}, \quad z = e^{i\theta},$$

where $\lambda > 0$, $d\theta/2\pi$ is the normalized Lebesgue measure on $[0, 2\pi]$ and $d\mu(\theta)$ is a measure belonging to the Bernstein–Szegő class (see [8]).

In [2] we have obtained a family of orthogonal polynomials with respect to a Sobolev inner product as above when $d\mu(\theta)$ is a particular Bernstein–Szegő measure, that is, $d\mu(\theta) = d\theta/(2\pi|e^{i\theta} - \alpha|^2)$. In that paper we have done a detailed study of the corresponding orthogonal polynomials, including some properties concerning asymptotic behavior and distribution of zeros.

The analysis of the results obtained there, and techniques used in the proofs lead us to think that most of the results may be generalized when $d\mu(\theta)$ belongs to the Bernstein–Szegő class. Indeed we believe that the behavior strongly depends on the fact that the Lebesgue measure is in the component of the inner product where the derivatives appear.

On the real line an important fact when the measures are chosen for the Sobolev inner products is their coherence (see [9, 13]). Although this study is in an initial step on the unit circle (see [6]), we think that our pair of measures could play an important role in this direction.

The organization of the paper is as follows. Section 2 is devoted to study the algebraic properties of the Sobolev polynomials and the main result is Theorem 2, in which the linear difference equation satisfied by the Sobolev polynomials is obtained. In Section 3 we deduce, from the difference equation satisfied by the coefficients, asymptotics for the Sobolev polynomials, and the main result is Theorem 7 concerning the relative asymptotics of a circle off radius less than one. As a consequence we also obtain, under very natural conditions, the limiting distribution of zeros.

2 THE BERNSTEIN-SZEGÖ-LEBESGUE SOBOLEV INNER PRODUCT

Let us consider a measure μ absolutely continuous with respect to the normalized Lebesgue measure such that the Radon-Nikodym derivative is the inverse of a positive trigonometric polynomial $d\mu(\theta) = d\theta/(2\pi r_h(\theta))$, with

$$r_h(\theta) = \sum_{l=0}^h (a_l \cos(l\theta) + b_l \sin(l\theta)), \quad r_h(\theta) > 0 \quad \forall \theta \in [0, 2\pi].$$

In this case F       and Riesz have shown that there exists a unique polynomial $A_h(z)$ such that $\deg A_h(z) = h$, $A_h(0) > 0$, $A_h(z) \neq 0$ for $|z| \leq 1$ and $r_h(\theta) = |A_h(e^{i\theta})|^2$ (see [8]). Besides, the sequence $\{\varphi_n\}$ defined by $\varphi_n(z) = z^{n-h} A_h^*(z)$ $\forall n \geq h$ is orthonormal with respect to the measure $d\mu(\theta) = d\theta/(2\pi |A_h(e^{i\theta})|^2)$. We recall that the $*$ -operator is defined in the space of polynomials $\mathbb{P}$ by $P^*(z) = z^n \bar{P}(z^{-1})$ for $P \in \mathbb{P}$, with $\deg P = n$ (see [7]).

We define the Bernstein-Szeg  -Lebesgue Sobolev inner product as follows:

$$\langle P, Q \rangle_s = \langle P, Q \rangle_\mu + \frac{1}{\lambda} \langle P', Q' \rangle_\theta,$$

that is,

$$\langle P, Q \rangle_s = \int_0^{2\pi} P(e^{i\theta}) \overline{Q(e^{i\theta})} \frac{d\theta}{2\pi |A_h(e^{i\theta})|^2} + \frac{1}{\lambda} \int_0^{2\pi} P'(e^{i\theta}) \overline{Q'(e^{i\theta})} \frac{d\theta}{2\pi}, \quad (1)$$

with $z = e^{i\theta}$, $P, Q \in \mathbb{P}$ and $\lambda > 0$, and we denote by $\{\tilde{\Phi}_n\}$ the monic orthogonal polynomial sequence (MOPS) with respect to $\langle \cdot, \cdot \rangle_s$.

In [3], by using the extremal property of the norms of the monic orthogonal polynomials we have obtained, in a more general situation, the next result about the Sobolev norm of the sequence $\tilde{\Phi}_n$.

THEOREM 1 *Let $\{\tilde{\Phi}_n\}$ be the MOPS related to $\langle \cdot, \cdot \rangle_s$. Then*

(i)

$$\lim_{n \rightarrow \infty} \frac{\|\tilde{\Phi}_n\|_s^2}{n^2} = \frac{1}{\lambda}.$$

(ii) *There exists $N \in \mathbb{N}$ such that $\{\|\tilde{\Phi}_n\|_s^2\}_{n \geq N}$ is an increasing sequence.*

Proof See [3].

THEOREM 2 Let $\{\tilde{\Phi}_n\}$ be the MOPS with respect to $\langle \cdot, \cdot \rangle_s$. Then there exist h sequences $\{\alpha_{n,n-i}\}_{n \geq h}$ ($i = 1, \dots, h$), such that

$$\varphi_n(z) = a_h \tilde{\Phi}_n(z) + \alpha_{n,n-1} \tilde{\Phi}_{n-1}(z) + \dots + \alpha_{n,n-h} \tilde{\Phi}_{n-h}(z) \quad n \geq h, \quad (2)$$

where a_h is the leading coefficient of the polynomial $A_h^*(z) = \sum_{k=0}^h a_k z^k$.
Moreover,

$$\lim_{n \rightarrow \infty} \alpha_{n,n-i} = a_{h-i} \quad (i = 1, \dots, h). \quad (3)$$

Proof Since $\{\varphi_n\}$ is orthogonal with respect to measure μ , then $\langle \varphi_n(z), z^k \rangle_\mu = 0$ for $k = 0, \dots, n-1$.

On the other hand, for $n \geq h$, $\varphi_n'(z) = (n-h)z^{n-h-1}A_h^*(z) + z^{n-h}(A_h^*(z))'$, and therefore $\langle \varphi_n'(z), z^k \rangle_\theta = 0$ for $k = 0, \dots, n-h-2$.

Then $\langle \varphi_n(z), z^k \rangle_s = 0$ for $k = 0, \dots, n-h-1$ with $n \geq h$ and so there exist $(h+1)$ sequences $\{\alpha_{n,n-i}\}_{n \geq h}$ ($i = 0, \dots, h$), such that $\varphi_n(z) = \sum_{i=0}^h \alpha_{n,n-i} \tilde{\Phi}_{n-i}(z)$.

In particular, comparing the leading coefficient we get $\alpha_{n,n} = a_h \forall n$.

From (2), $\langle \varphi_n, \tilde{\Phi}_{n-i} \rangle_s = \alpha_{n,n-i} \|\tilde{\Phi}_{n-i}\|_s^2$, which yields for $i = 1, \dots, h$,

$$\begin{aligned} \alpha_{n,n-i} &= \frac{\langle \varphi_n, \tilde{\Phi}_{n-i} \rangle_s}{\|\tilde{\Phi}_{n-i}\|_s^2} = \frac{\langle \sum_{l=0}^i a_{h-l} z^{n-l}, \tilde{\Phi}_{n-i} \rangle_s}{\|\tilde{\Phi}_{n-i}\|_s^2} \\ &= a_{h-i} \frac{\langle z^{n-i}, \tilde{\Phi}_{n-i} \rangle_s}{\|\tilde{\Phi}_{n-i}\|_s^2} + \frac{\langle \sum_{l=0}^{i-1} a_{h-l} z^{n-l}, \tilde{\Phi}_{n-i} \rangle_s}{\|\tilde{\Phi}_{n-i}\|_s^2} \\ &= a_{h-i} + \sum_{l=0}^{i-1} \frac{\langle a_{h-l} z^{n-l}, \tilde{\Phi}_{n-i} \rangle_\mu}{\|\tilde{\Phi}_{n-i}\|_s^2}. \end{aligned} \quad (4)$$

Thus,

$$\alpha_{n,n-i} - a_{h-i} = \sum_{l=0}^{i-1} \frac{\langle a_{h-l} z^{n-l}, \tilde{\Phi}_{n-i} \rangle_\mu}{\|\tilde{\Phi}_{n-i}\|_s^2}. \quad (5)$$

On the other hand,

$$\frac{|\langle a_{h-l} z^{n-l}, \tilde{\Phi}_{n-i} \rangle_\mu|^2}{\|\tilde{\Phi}_{n-i}\|_s^4} \leq \frac{|a_{h-l}|^2 \|z^{n-l}\|_\mu^2 \|\tilde{\Phi}_{n-i}\|_\mu^2}{\|\tilde{\Phi}_{n-i}\|_s^4} \leq \frac{|a_{h-l}|^2 c_0}{\|\tilde{\Phi}_{n-i}\|_s^2}, \quad (6)$$

where we denote by $c_0 = \langle 1, 1 \rangle_\mu$.

Then, taking into account Theorem 1, we obtain

$$\lim_{n \rightarrow \infty} \frac{|\langle a_{h-l} z^{n-l}, \tilde{\Phi}_{n-i} \rangle_\mu|}{\|\tilde{\Phi}_{n-i}\|_s^2} = 0$$

and therefore $\lim_{n \rightarrow \infty} \alpha_{n,n-i} = a_{h-i}$ for $i = 1, \dots, h$.

COROLLARY 1 *The MOPS related to $\langle \cdot, \cdot \rangle_s \{ \tilde{\Phi}_n \}$ verifies the following $(h+2)$ -term recurrence relation*

$$\begin{aligned} \tilde{\Phi}_{n+1}(z) &= \left(z - \frac{\alpha_{n+1,n}}{a_h} \right) \tilde{\Phi}_n(z) + \sum_{i=1}^{h-1} \left(\frac{\alpha_{n,n-i} z - \alpha_{n+1,n-i}}{a_h} \right) \tilde{\Phi}_{n-i}(z) \\ &\quad + \frac{\alpha_{n,n-h}}{a_h} z \tilde{\Phi}_{n-h}(z) \quad n \geq h. \end{aligned} \quad (7)$$

Proof If we write (2) for $n+1$ and take into account that $\varphi_{n+1}(z) = z\varphi_n(z)$ for $n \geq h$, we get

$$z\varphi_n(z) = a_h \tilde{\Phi}_{n+1}(z) + \sum_{i=0}^{h-1} \alpha_{n+1,n-i} \tilde{\Phi}_{n-i}(z). \quad (8)$$

From (2) and (8) we eliminate $\varphi_n(z)$ and obtain (7).

COROLLARY 2 *Let $\alpha_{n,n-i}$ ($i = 1, \dots, h$) be the coefficients in relation (2). Then there exists $N \in \mathbb{N}$ such that*

$$|\alpha_{n,n-i}| < \|A_h^*(z)\|_1 = \sum_{j=0}^h |a_j| \quad \text{for } n \geq N \geq h \quad (i = 1, \dots, h).$$

Proof From (4) and (6)

$$|\alpha_{n,n-i}| \leq |a_{h-i}| + \sum_{l=0}^{i-1} \frac{|a_{h-l}|c_0^{1/2}}{\|\tilde{\Phi}_{n-i}\|_s}.$$

Since

$$\lim_{n \rightarrow \infty} \frac{1}{\|\tilde{\Phi}_{n-i}\|_s} = 0,$$

there exists $N \in \mathbb{N}$ such that $\forall n \geq N$,

$$\frac{1}{\|\tilde{\Phi}_{n-i}\|_s} < \frac{1}{c_0^{1/2}}$$

and thus $|\alpha_{n,n-i}| < \sum_{l=0}^i |a_{h-l}| \leq \|A_h^*(z)\|_1$.

By using the extremal properties of the norms of the monic orthogonal polynomials, we also have obtained in [3] the following result that we recall:

THEOREM 3 *It holds that*

$$\|\tilde{\Phi}_n - z^n\|_s^2 \leq c_0 - \|\tilde{\Phi}_n\|_\mu^2. \quad (9)$$

Proof See [3].

Next we obtain, as a consequence, a result about the behavior of the coefficients of the sequence $\{\tilde{\Phi}_n\}$.

THEOREM 4 *Let $\{\tilde{\Phi}_n\}$ be the MOPS related to $\langle \cdot, \cdot \rangle_s$. If $\tilde{\Phi}_n(z) = z^n + \sum_{k=0}^{n-1} A_{n,k} z^k$, then it holds that*

$$|A_{n,k}| \leq \frac{\lambda c_0^2}{k^2} \quad k = 1, \dots, n-1. \quad (10)$$

Proof For $1 \leq k < n$, it holds

$$0 = \langle \tilde{\Phi}_n, z^k \rangle_s = \langle \tilde{\Phi}_n, z^k \rangle_\mu + \frac{1}{\lambda} \langle \tilde{\Phi}_n', k z^{k-1} \rangle_\theta,$$

which implies

$$\frac{1}{\lambda} \langle \tilde{\Phi}'_n, k z^{k-1} \rangle_\theta = \frac{1}{\lambda} A_{n,k} k^2 = -\langle \tilde{\Phi}_n, z^k \rangle_\mu.$$

Therefore

$$\frac{1}{\lambda} |A_{n,k}| k^2 \leq \|\tilde{\Phi}_n\|_\mu \|z^k\|_\mu,$$

and then, taking into account (9) we deduce (10).

Remark 1 From Theorem 4 we also deduce the following property of the coefficients of $\tilde{\Phi}_n$:

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \text{ such that for } n \geq N \quad |A_{n,k}| < \epsilon \quad \text{for } k = N, \dots, n-1.$$

Finally, we recall the following facts. Since measure μ verifies the Szegő's condition (see [7]), and the series $\sum_{n=0}^{\infty} |c_n|$ converges, with $c_n = \langle z^n, 1 \rangle_\mu$, then we deduce (see [3, Theorem 3]), that

$$\lim_{n \rightarrow \infty} \|\tilde{\Phi}_n - z^n\|_s = 0, \quad (11)$$

which means that for n large enough, $\tilde{\Phi}_n$ can be approximated by z^n in the Sobolev norm, and

$$\lim_{n \rightarrow \infty} \|\tilde{\Phi}_n\|_\mu = c_0. \quad (12)$$

3 ASYMPTOTIC PROPERTIES AND DISTRIBUTION OF ZEROS

THEOREM 5 We denote z_i ($i = 1, \dots, h$) the zeros of $A_h^*(z)$. Let ρ_1 such that $\max_{i=1, \dots, h} \{|z_i|\} < \rho_1 < 1$. Then there exist $M > 0$ and $N \in \mathbb{N}$ such that for $n \geq N$

$$|A_{n,k}| \leq \rho_1^{n-k-h} \frac{M}{k^2} \quad (k \geq 1) \quad (13)$$

and

$$|A_{n,0}| \leq \rho_1^{n-h} M. \quad (14)$$

Proof Let us consider the recurrence relation (2). If $\tilde{\Phi}_n(z) = z^n + \sum_{k=0}^{n-1} A_{n,k} z^k$, then we deduce from (2) the following linear homogeneous difference equation:

$$0 = a_h A_{n,k} + \alpha_{n,n-1} A_{n-1,k} + \cdots + \alpha_{n,n-h} A_{n-h,k} \quad (k = 0, \dots, n-h-1), \quad (15)$$

which is equivalent to the system

$$\begin{pmatrix} A_{n-h+1,k} \\ \vdots \\ A_{n,k} \end{pmatrix} = \mathbf{A}_n \begin{pmatrix} A_{n-h,k} \\ \vdots \\ A_{n-1,k} \end{pmatrix}, \quad (16)$$

with

$$\mathbf{A}_n = \begin{pmatrix} 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ \frac{-\alpha_{n,n-h}}{a_h} & \frac{-\alpha_{n,n-h+1}}{a_h} & \cdots & \frac{-\alpha_{n,n-2}}{a_h} & \frac{-\alpha_{n,n-1}}{a_h} \end{pmatrix}.$$

Since $\lim_{n \rightarrow \infty} -\alpha_{n,n-i} = -a_{h-i}$ ($i = 1, \dots, h$), then the sequence of matrices $\{\mathbf{A}_n\}$ converges to the matrix

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ \frac{-a_0}{a_h} & \frac{-a_1}{a_h} & \cdots & \frac{-a_{h-2}}{a_h} & \frac{-a_{h-1}}{a_h} \end{pmatrix},$$

such that its spectral radius $\rho(\mathbf{A}) = \max_{i=1, \dots, h} \{|z_i|\} < 1$.

Since $\rho(\mathbf{A}) = \inf\{\|\mathbf{A}\| : \text{with } \|\cdot\| \text{ induced by a vector norm}\}$; if we take ρ_1 such that $\rho(\mathbf{A}) < \rho_1 < 1$, then there exists a norm in $\mathbb{C}^h$ such that the corresponding matrix norm verifies $\rho(\mathbf{A}) < \|\mathbf{A}\| < \rho_1 < 1$.

The convergence of $\mathbf{A}_n$ to $\mathbf{A}$ implies for n large enough $\|\mathbf{A}_n\| < \rho_1$. For simplicity and without loss of generality we may assume that $\|\mathbf{A}_n\| < \rho_1 \forall n$. Hence, if we take $k < n - h$ and apply (16) recursively, we deduce

$$\begin{aligned} \begin{pmatrix} A_{n-h+1,k} \\ \vdots \\ A_{n,k} \end{pmatrix} &= \mathbf{A}_n \begin{pmatrix} A_{n-h,k} \\ \vdots \\ A_{n-1,k} \end{pmatrix} = \mathbf{A}_n \mathbf{A}_{n-1} \begin{pmatrix} A_{n-h-1,k} \\ \vdots \\ A_{n-2,k} \end{pmatrix} \\ &= \mathbf{A}_n \mathbf{A}_{n-1} \cdots \mathbf{A}_{k+h+1} \begin{pmatrix} A_{k+1,k} \\ \vdots \\ A_{k+h,k} \end{pmatrix}, \end{aligned}$$

and therefore

$$\left\| \begin{pmatrix} A_{n-h+1,k} \\ \vdots \\ A_{n,k} \end{pmatrix} \right\| \leq \rho_1^{n-k-h} \left\| \begin{pmatrix} A_{k+1,k} \\ \vdots \\ A_{k+h,k} \end{pmatrix} \right\|. \quad (17)$$

Hence, combining (10) with the fact that in $\mathbb{C}^h$ our norm is equivalent to $\|\cdot\|_\infty$, we deduce from (17)

$$c|A_{n,k}| \leq \rho_1^{n-k-h} \frac{C\lambda c_0^2}{k^2} \quad (k \geq 1),$$

that is

$$|A_{n,k}| \leq \rho_1^{n-k-h} \frac{M}{k^2} \quad (k \geq 1).$$

Similarly, we can obtain that $|A_{n,0}| \leq \rho_1^{n-h} M$.

In the case when the zeros of A_h^* are distinct, we can improve this last result obtaining better bounds for the coefficients.

THEOREM 6 *Let us assume that the zeros z_i ($i=1, \dots, h$) of A_h^* are different and let $|z_1| = \max_{i=1, \dots, h} \{|z_i|\}$. Then there exist $M_1 > 0$ and*

$M_2 > 0$ such that

$$|A_{n,k}| \leq |z_1|^{n-k-h} \frac{M_1}{k^2} \quad (k \geq 1), \quad (18)$$

and

$$|A_{n,0}| \leq |z_1|^{n-h} M_2. \quad (19)$$

for $n > k + h$.

Proof Since the zeros of A_h^* are different, we have that the matrix $\mathbf{A}$ in the preceding theorem has different eigenvalues and, therefore we can choose a matrix norm such that $\|\mathbf{A}\| = \rho(\mathbf{A}) = |z_1|$.

Let $\mathbf{A}_n$ be the matrix in Eq. (16). It is clear that $\mathbf{A}_n$ can be written as $\mathbf{A}_n = \mathbf{A} + \mathbf{B}_n$, with

$$\mathbf{B}_n = \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 \\ \frac{-\alpha_{n,n-h} + a_0}{a_h} & \frac{-\alpha_{n,n-h+1} + a_1}{a_h} & \cdots & \frac{-\alpha_{n,n-2} + a_{h-2}}{a_h} & \frac{-\alpha_{n,n-1} + a_{h-1}}{a_h} \end{pmatrix}$$

and therefore

$$\|\mathbf{A}_n\| \leq |z_1| + \|\mathbf{B}_n\| \leq |z_1| + C \max_{i=1,\dots,h} \left\{ \frac{|\alpha_{n,n-i} - a_{h-i}|}{|a_h|} \right\}.$$

From (5) and taking into account (9)

$$\begin{aligned} |\alpha_{n,n-i} - a_{h-i}| &\leq \frac{\sum_{l=0}^{i-1} |\langle a_{h-l} z^{n-l}, \tilde{\Phi}_{n-i} \rangle_\mu|}{\|\tilde{\Phi}_{n-i}\|_s^2} \\ &\leq \frac{\sum_{l=0}^{i-1} |a_{h-l}| c_0^2}{\|\tilde{\Phi}_{n-i}\|_s^2} \leq \lambda c_0^2 \frac{\sum_{l=0}^{i-1} |a_{h-l}|}{(n-i)^2}, \end{aligned}$$

where the last inequality follows from the fact that $\|\tilde{\Phi}_n\|_s^2 \geq n^2/\lambda$.

Thus,

$$\max_{i=1,\dots,h} \left\{ \frac{|\alpha_{n,n-i} - a_{h-i}|}{|a_h|} \right\} \leq \frac{\lambda c_0^2}{|a_h|} \frac{|a_h| + \dots + |a_1|}{(n-h)^2}$$

and so

$$\|A_n\| \leq |z_1| \left(1 + \frac{\lambda T}{(n-h)^2} \right) \quad \text{with } T = \frac{C c_0^2 (|a_h| + \dots + |a_1|)}{|a_h| |z_1|}.$$

Now if we apply (16) recursively and take norms we obtain

$$\begin{aligned} & \left\| \begin{pmatrix} A_{n-h+1,k} \\ \vdots \\ A_{n,k} \end{pmatrix} \right\| \\ & \leq |z_1|^{n-k-h} \left(1 + \frac{\lambda T}{(n-h)^2} \right) \cdots \left(1 + \frac{\lambda T}{(k+1)^2} \right) \left\| \begin{pmatrix} A_{k+1,k} \\ \vdots \\ A_{k+h,k} \end{pmatrix} \right\| \end{aligned}$$

from which we deduce, by using (10)

$$c_1 |A_{n,k}| \leq |z_1|^{n-k-h} \left(1 + \frac{\lambda T}{(n-h)^2} \right) \cdots \left(1 + \frac{\lambda T}{(k+1)^2} \right) C_1 \frac{\lambda c_0^2}{k^2},$$

that is

$$|A_{n,k}| \leq \frac{|z_1|^{n-k-h}}{k^2} M_1 \quad \text{with } M_1 = \frac{C_1 \lambda c_0^2}{c_1} \prod_{i=1}^{\infty} \left(1 + \frac{\lambda T}{i^2} \right) \quad (k \geq 1).$$

In the same way we obtain the case $k=0$.

We note that taking λ small we can do M_1 as small as we want.

By using some well-known results about difference equations we can improve the bounds in Theorems 5 and 6, in the case when the zeros of A_h^* are distinct in moduli.

By applying Poincaré's Theorem (see [1]), to the difference equation (15), we deduce that every non-trivial solution $A_{n,k}$ satisfies that $\lim_{n \rightarrow \infty} A_{n+1,k}/A_{n,k}$ is equal to one of the zeros of the characteristic polynomial $a_h \lambda^h + \dots + a_1 \lambda + a_0 = A_h^*(\lambda)$, provided all these zeros are distinct in modulus. Now we assume the roots of A_h^* have distinct moduli, and we arrange them so that $|z_1| > \dots > |z_h|$. Moreover, Perron's theorem (see [1]) allows us to obtain a fundamental set of solutions of (15) $v_i(n)$, $i = 1, \dots, h$ with the property $v_i(n) = z_i^n(1 + o_i(1))$.

Therefore, for each k , there exists $i \in \{1, \dots, h\}$ such that $\lim_{n \rightarrow \infty} A_{n,k}/z_i^n = L_k \neq 0$. Let

$$|z_l| = \max \left\{ |z_i| : \text{for some } k \lim_{n \rightarrow \infty} \frac{A_{n,k}}{z_i^n} \neq 0 \right\}.$$

In this situation we can state the following:

COROLLARY 3 *If*

$$|z_l| = \max \left\{ |z_i| : \text{for some } k \lim_{n \rightarrow \infty} \frac{A_{n,k}}{z_i^n} \neq 0 \right\}$$

and $|z_i| \neq |z_j|$ for $i \neq j$, then for $\epsilon > 0$ there exist $M > 0$ and $N_1 \in \mathbb{N}$ such that for $n \geq N_1$

$$|A_{n,k}| \leq (|z_l| + \epsilon)^{n-k-h} \frac{M}{k^2} \quad (k \geq 1) \quad (20)$$

and

$$|A_{n,0}| \leq (|z_l| + \epsilon)^{n-h} M. \quad (21)$$

Proof Proceeding in the same way as in the proof of Theorem 5, from (16) we deduce (17), substituting ρ_1 by $|z_l| + \epsilon$ with $\epsilon > 0$. Take into account that the spectral radius of A restricted to the subspace generated by the eigenvectors associated with the eigenvalues $|z_l|, \dots, |z_h|$ is $|z_l|$. Then for n large enough we get relations (20) and (21).

In the results that come next we assume again z_l is chosen as above, and when $l \neq 1$ we also suppose that the roots of A_h^* have distinct moduli.

THEOREM 7 Let $\{\tilde{\Phi}_n\}$ be the MOPS related to $\langle \cdot, \cdot \rangle_s$. Let

$$|z_l| = \max \left\{ |z_l| : \text{for some } k \lim_{n \rightarrow \infty} \frac{A_{n,k}}{z_l^n} \neq 0 \right\}$$

and if $l \neq 1$, assume $|z_l| \neq |z_j|$ for $i \neq j$. Then

$$\lim_{n \rightarrow \infty} \frac{\tilde{\Phi}_n(z)}{z^n} = 1 \quad \text{uniformly on compact subsets of } |z| > |z_l|.$$

Proof In [3], in a more general situation, we have obtained the convergence uniformly on compact subsets of $|z| > 1$. Here we improve this result obtaining convergence on and inside the unit disk.

For simplicity, we assume for the proof, $|z_l| = |z_1|$. Let z such that $1 \geq |z| > |z_1|$. Then there exists ρ_1 such that $|z_l| < \rho_1 < |z|$. Let

$$S = \frac{M}{\rho_1^h} \sum_{n=0}^{\infty} \left(\frac{\rho_1}{|z|} \right)^n$$

with M defined as in Theorem 5.

For a fixed $\epsilon > 0$, let us take $N \in \mathbb{N}$ such that $1/N^2 < \epsilon/3S$. Also we take m such that

$$\frac{M}{\rho_1^h} \left(\frac{\rho_1}{|z|} \right)^m < \frac{\epsilon}{3N}$$

and p verifying that

$$\frac{\lambda c_0^2 h}{|z|^h (p-h)^2} < \frac{\epsilon}{3}.$$

Then, for $n \geq \max\{N+h+1, m+N-1, p\}$ and n large enough such that (13) and (14) hold, we have that

$$\begin{aligned} \left| \frac{\tilde{\Phi}_n(z)}{z^n} - 1 \right| &= \left| \sum_{k=0}^{n-1} \frac{A_{n,k}}{z^{n-k}} \right| \\ &\leq \sum_{k=n-h}^{n-1} \frac{|A_{n,k}|}{|z|^{n-k}} + \sum_{k=N}^{n-h-1} \frac{|A_{n,k}|}{|z|^{n-k}} + \sum_{k=0}^{N-1} \frac{|A_{n,k}|}{|z|^{n-k}}. \end{aligned}$$

Applying (10)

$$\sum_{k=n-h}^{n-1} \frac{|A_{n,k}|}{|z|^{n-k}} < \sum_{k=n-h}^{n-1} \frac{\lambda c_0^2}{|z|^{n-k} k^2} < \frac{\lambda c_0^2 h}{|z|^h (n-h)^2} < \frac{\epsilon}{3}.$$

Applying (13) and (14) we obtain

$$\begin{aligned} \sum_{k=N}^{n-h-1} \frac{|A_{n,k}|}{|z|^{n-k}} &< \sum_{k=N}^{n-h-1} \frac{\rho_1^{n-k-h} M}{|z|^{n-k} k^2} = \sum_{k=N}^{n-h-1} \frac{M}{\rho_1^h} \left(\frac{\rho_1}{|z|} \right)^{n-k} \frac{1}{k^2} \\ &< \frac{\epsilon}{3S} \sum_{k=N}^{n-h-1} \frac{M}{\rho_1^h} \left(\frac{\rho_1}{|z|} \right)^{n-k} < \frac{\epsilon}{3}, \end{aligned}$$

and

$$\begin{aligned} \sum_{k=0}^{N-1} \frac{|A_{n,k}|}{|z|^{n-k}} &= \sum_{k=1}^{N-1} \frac{|A_{n,k}|}{|z|^{n-k}} + \frac{|A_{n,0}|}{|z|^n} \\ &\leq \sum_{k=1}^{N-1} \frac{\rho_1^{n-k-h} M}{|z|^{n-k} k^2} + \frac{\rho_1^{n-h} M}{|z|^n} \\ &\leq \sum_{k=0}^{N-1} \frac{M}{\rho_1^h} \left(\frac{\rho_1}{|z|} \right)^{n-k} < N \frac{\epsilon}{3N} = \frac{\epsilon}{3}. \end{aligned}$$

Therefore for $\epsilon > 0$ there exists k such that

$$\forall n \geq k, \quad \left| \frac{\tilde{\Phi}_n(z)}{z^n} - 1 \right| < \epsilon \quad \forall z, \quad |z_1| < |z|.$$

The proof, for the general case, can be obtained in the same way taking into account Corollary 3 and using Eqs. (20) and (21), instead of (13) and (14).

COROLLARY 4 *Under assumptions of Theorem 7*

$$\lim_{n \rightarrow \infty} \frac{\tilde{\Phi}_n(z)}{\varphi_n(z)} = \frac{z^h}{A_h^*(z)} \quad \text{uniformly on compact subsets of } |z| > |z_l|. \quad (22)$$

Proof Since

$$\frac{\tilde{\Phi}_n(z)}{\varphi_n(z)} = \frac{\tilde{\Phi}_n(z)}{z^{n-h} A_h^*(z)} = \frac{\tilde{\Phi}_n(z)}{z^n} \frac{z^h}{A_h^*(z)},$$

then applying Theorem 7 we have the result.

THEOREM 8 *The roots of $\tilde{\Phi}_n$ are in the disk $|z| < 1 + \max\{\lambda c_0^2, P\}$.*

Proof We apply a result due to Cauchy (see [12]) which says that the zeros of $\tilde{\Phi}_n$ are in the annulus

$$\frac{|A_{n,0}|}{|A_{n,0}| + L'} < |z| < 1 + L$$

with $L = \max_{k=0, \dots, n-1} \{|A_{n,k}|\}$ and $L' = \max_{k=1, \dots, n-1} \{1, |A_{n,k}|\}$.

From Theorem 5, the sequence $\{A_{n,0}\}$ is bounded, that is, $|A_{n,0}| \leq P$, and from Theorem 4, $|A_{n,k}| \leq \lambda c_0^2/k^2$ for $k=1, \dots, n-1$. Then $L \leq \max\{\lambda c_0^2, P\}$.

COROLLARY 5 *Under assumptions of Theorem 7, if $\epsilon > 0$, then for n large enough the zeros of $\tilde{\Phi}_n$ are in $|z| < |z_l| + \epsilon$.*

Proof The result follows immediately if we take into account Theorem 8, and apply Hurwitz's theorem (see [5]), in Theorem 7.

THEOREM 9 *Let $\{\tilde{\Phi}_n\}$ be the MOPS related to $\langle \cdot, \cdot \rangle_s$. Then*

$$\lim_{n \rightarrow \infty} \tilde{\Phi}_n(z) = 0 \quad \text{uniformly on compact subsets of } |z| < 1.$$

Proof We use the lemma due to Maté and Nevai (see [11]) concerning the asymptotic behavior of solutions of linear recurrence equations. For a fixed z with $|z| < 1$ we consider the recurrence equation (2). We take into account that the characteristic polynomial $A_h^*(z)$ has no zeros of absolute value 1 and it is easy to see that $\{\tilde{\Phi}_n(z)\}$ is bounded.

Besides, since

$$\lim_{n \rightarrow \infty} \frac{\varphi_n(z)}{n^{-2}} = \lim_{n \rightarrow \infty} \frac{z^{n-h} A_h^*(z)}{n^{-2}} = 0, \quad \text{then} \quad \lim_{n \rightarrow \infty} \frac{\tilde{\Phi}_n(z)}{n^{-2}} = 0.$$

Therefore $\lim_{n \rightarrow \infty} \tilde{\Phi}_n(z) = 0$.

On the other hand, if we consider the region $D = \{z: |z| < 1\}$ it holds that the sequence $\{\tilde{\Phi}_n(z)\} \subset H(D)$ is locally bounded and $A = \{z \in D: \lim_{n \rightarrow \infty} \tilde{\Phi}_n(z) = 0\}$ has a limit point in D . Therefore,

applying Vitali's theorem (see [5]) we deduce that $\{\tilde{\Phi}_n(z)\}$ converges to 0 uniformly on compact subsets of D .

Finally, under a suitable assumption for the behavior of the parameter λ in the Sobolev product (1), we obtain the distribution of zeros for the sequence $\{\tilde{\Phi}_n\}$.

THEOREM 10 *Assume the zeros z_i of A_h^* verify that $|z_1| > |z_j| \forall j \neq 1$. Then there exists λ_0 such that for $0 < \lambda \leq \lambda_0$ the corresponding MOPS $\{\tilde{\Phi}_n\}$ verifies*

$$\lim_{n \rightarrow \infty} \sqrt[n]{|\tilde{\Phi}_n(0)|} = |z_1|.$$

Proof Since

$$0 = \langle \tilde{\Phi}_n, 1 \rangle_s = \left\langle z^n + \sum_{k=1}^{n-1} A_{n,k} z^k + \tilde{\Phi}_n(0), 1 \right\rangle_s,$$

then

$$\tilde{\Phi}_n(0) = \frac{-\left\langle z^n + \sum_{k=1}^{n-1} A_{n,k} z^k, 1 \right\rangle_s}{\langle 1, 1 \rangle_s} = \frac{-c_n - \sum_{k=1}^{n-1} A_{n,k} c_k}{c_0}.$$

Now, if we use (10) and (18), and take into account that the moments for the measure μ are $c_n = z_1^n(d + o(1))$, we get

$$\begin{aligned} c_0 |\tilde{\Phi}_n(0)| &= \left| c_n + \sum_{k=1}^{n-1} A_{n,k} c_k \right| \geq |c_n| - \sum_{k=1}^{n-1} |A_{n,k}| |c_k| \\ &\geq |z_1|^n |d + o(1)| - \sum_{k=1}^{n-h-1} \frac{|z_1|^{n-k-h}}{k^2} M_1 |z_1|^k |d + o(1)| \\ &\quad - \sum_{k=n-h}^{n-1} \frac{\lambda c_0^2}{k^2} |z_1|^k |d + o(1)| \\ &\geq |z_1|^n \left(|d + o(1)| - \frac{M_1}{|z_1|^h} \sum_{k=1}^{n-h-1} \frac{|d + o(1)|}{k^2} \right) - \frac{\lambda c_0^2 h}{(n-h)^2} |z_1|^{n-h} |d + o(1)| \\ &\geq |z_1|^n \left(|d + o(1)| - \frac{M_1}{|z_1|^h} \sum_{k=1}^{\infty} \frac{|d + o(1)|}{k^2} - \frac{\lambda c_0^2 h}{(n-h)^2 |z_1|^h} |d + o(1)| \right). \end{aligned}$$

Since $\sum_{k=1}^{\infty} |d + o(1)|/k^2 < \infty$, we can choose λ_0 such that $M_1(\lambda_0) = M_1$ verifies

$$\lim_{n \rightarrow \infty} \left(|d + o(1)| - \frac{M_1}{|z_1|^h} \sum_{k=1}^{\infty} \frac{|d + o(1)|}{k^2} - \frac{\lambda_0 c_0^2 h}{(n-h)^2 |z_1|^h} |d + o(1)| \right) > 0.$$

Therefore,

$$\lim_{n \rightarrow \infty} \frac{|\tilde{\Phi}_n(0)|}{|z_1|^n} \neq 0,$$

which implies $\lim_{n \rightarrow \infty} \sqrt[n]{|\tilde{\Phi}_n(0)|} = |z_1|$.

Finally, under a very natural assumption on the behavior of $\{\tilde{\Phi}_n(0)\}$, like in this last situation, we can obtain weak distribution of zeros:

THEOREM 11 *Let $|z_l| = \max\{|z_i|: \text{for some } k \lim_{n \rightarrow \infty} A_{n,k}/z_i^n \neq 0\}$ and if $l \neq 1$, assume $|z_l| \neq |z_j|$ for $i \neq j$. If $\lim_{n \rightarrow \infty} \sqrt[n]{|\tilde{\Phi}_n(0)|} = |z_l|$, then in the weak* topology*

$$\lim_{n \rightarrow \infty} \nu(\tilde{\Phi}_n) = \nu_{|z_l|},$$

where we denote by $\nu(\tilde{\Phi}_n)$ the discrete unit measure defined on Borel sets in $\mathbb{C}$ having mass $1/n$ at each zero of $\tilde{\Phi}_n$, and by $\nu_{|z_l|}$ the arc-measure $d\theta/(2\pi|z_l|)$ on the circle $C_{|z_l|}$, that is, the equilibrium distribution $C_{|z_l|}$.

Proof For n large enough we consider the following monic polynomials of degree n , $\Psi_n(z) =: \tilde{\Phi}_n^*(z)/\tilde{\Phi}_n(0)$. Let us consider the circle $C_R = \{z: |z| = R\}$ with capacity R . For $R = 1/|z_l|$, next we prove the following two conditions:

- (i) $\overline{\lim}_{n \rightarrow \infty} \sqrt[n]{\|\Psi_n\|_{C_R}} \leq R$, where $\|\cdot\|_K$ denotes the sup norm over the compact K . Indeed,

$$\|\tilde{\Phi}_n\|_{C_1} \leq 1 + \sum_{k=0}^{n-1} |A_{n,k}| \leq 2 + \sum_{k=1}^{n-1} \frac{\lambda c_0^2}{k^2} \leq L.$$

Then

$$\|\tilde{\Phi}_n^*\|_{C_R} = R^n \|\tilde{\Phi}_n\|_{C_{1/R}} \leq L,$$

where the last inequality follows from Bernstein's lemma (see [15, Section 4.6]). Therefore

$$\begin{aligned} \overline{\lim}_{n \rightarrow \infty} \sqrt[n]{\|\Psi_n\|_{C_R}} &= \overline{\lim}_{n \rightarrow \infty} \sqrt[n]{\frac{\|\tilde{\Phi}_n^*\|_{C_R}}{|\tilde{\Phi}_n(0)|}} \\ &\leq \overline{\lim}_{n \rightarrow \infty} \sqrt[n]{\frac{L}{|\tilde{\Phi}_n(0)|}} = R. \end{aligned}$$

- (ii) For each closed set A contained in the bounded component of the outer boundary of C_R , $\lim_{n \rightarrow \infty} \nu(\Psi_n)(A) = 0$.

In our hypothesis, from Corollary 5, we obtain that for $\epsilon > 0$ and n large enough, the zeros of $\tilde{\Phi}_n$ are in $|z| < |z_l| + \epsilon$. Therefore the zeros of Ψ_n are outside the circle of radius $1/(|z_l| + \epsilon)$, and this implies $\lim_{n \rightarrow \infty} \nu(\Psi_n)(A) = 0$.

Therefore, from (i) and (ii) and applying Theorem 2.1 in [4], we obtain $\nu(\Psi_n)$ converges weakly to the equilibrium distribution of C_R , and thus $\nu(\tilde{\Phi}_n)$ converges weakly to the equilibrium distribution of $C_{|z_l|}$.

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